Steps for calculating trend values when n is odd:

i) Subtract the first year from all the years (x)

ii) Take the middle value (A)

iii) Find $u_i = x_i - A$

iv) Find u_i^2 and $u_i y_i$

Then use the normal equations:

$$\sum_{i=1}^{n} y_i = na + b\sum_{i=1}^{n} u_i$$

$$\sum_{i=1}^{n} u_i y_i = a\sum_{i=1}^{n} u_i + b\sum_{i=1}^{n} u_i^2$$

Find $a = \dfrac{\sum_{i=1}^{n} y_i}{n}$ and $b = \dfrac{\sum_{i=1}^{n} u_i y_i}{\sum_{i=1}^{n} u_i^2}$

Then the estimated equation of straight line is:

$y = a + b\,u = a + b\,(x - A)$

Example 7.7

Fit a straight line trend by the method of least squares for the following consumer price index numbers of the industrial workers.

Year	2010	2011	2012	2013	2014
Index number	166	177	198	221	225

Solution:

Year	Index Number	$X = x_i - 2010$	$u_i = X - A$ $= X - 2$	u_i^2	$u_i y_i$	Trend
2010	166	0	–2	4	–332	165
2011	177	1	–1	1	–177	181.2
2012	198	2	0	0	0	197.4
2013	221	3	1	1	221	213.6
2014	225	4	2	4	450	229.8
	$\sum_{i=1}^{5} y_i = 987$		$\sum_{i=1}^{5} u_i = 0$	$\sum_{i=1}^{5} u_i^2 = 10$	$\sum_{i=1}^{5} u_i y_i = 162$	

The equation of the straight line is $y = a + bx$

$= a + bu$ where $u = X - 2$

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